

AI coding overview

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Summary

This page is an overview of the choices you face when doing AI coding (qualitative causal coding with AI) inside the Causal Map app.

- For step-by-step app docs see [AI coding](#)
- For the broader argument about why we use AI as a low-level coder rather than as a black box, see [AI in evaluation actually](#)

[show your working!](#) and [Just add rigour Three do's and don'ts.](#)

Workflow

Someone arrives with a stack of documents and asks: can you code these? It depends first on various parallel setup decisions. After coding, you face a separate decision about how, or whether, to recode the labels.

Decisions before coding

Random-sample coding strategy with bigger corpora

We will usually test and improve the instruction(s) on just a small random sample of the material. It's important that it is random, so that you don't waste time fitting an instruction to just one type of material which is maybe not typical for the whole corpus of text. The Causal Map app allows you to [select random samples](#) overall or, even better, random samples from each of relevant groups like women and men.

If you have more than around 100 pages, work on a sample first. The app has features to take a random sample of sources, or a stratified random sample within source groups. The AI coding panel also offers a sample-only run (see [AI coding](#)). Try the sample, review, adjust your strategy, perhaps update the codebook, then run a larger sample. With 1000 pages you might code 100, adjust, code another 300 (deleting the first attempt), and if that looks good, finish with the remaining 700.

Using additional iterations

When [coding with AI](#), we often use one or more additional AI iterations, e.g.

- Accuracy: to check that the coded links have been coded correctly (and if necessary, then amend/delete them)
 - according to the original rules
 - according to new/additional criteria
- Coverage[±]: to check that links which should have been coded were indeed identified and if not to add them. Each additional iteration sends the original text and instructions and the results back to the AI, i.e. the "conversation" to date, along with additional specific instructions for this iteration. So if you add two additional iterations, the results will take altogether three times as long (and cost three times as much) as the same task without iterations.

We often find that our time is better spent improving the original coding instruction than adding additional iterations.

Improving the instructions

The most important trick of all is not just to apply a generic prompt / coding instruction, with or without additional iterations, but in initial testing to examine the results, work out specifically the reasons for any problems with accuracy or coverage, and to **tweak the instruction and try again**, repeating until you are satisfied. We usually do this before adding any additional iterations and then if the additional iteration(s) do improve the results, we check them as above to see if we can improve the *iteration* instruction.

When there is a lot of text to process (hundreds or thousands of pages of text) we will sometimes start off with a small sample as described above and then, once we are satisfied, check again on a larger sample before coding the entire corpus.

Codebook strategy: how free?

You can start with a full codebook, a partial codebook, or nothing.

A *forced* codebook restricts the model to your labels (for example from a theory of change). If a causal claim mentions a factor not in the codebook, no link is coded.

Most often you provide a codebook but allow the model to invent labels when nothing fits. This is harder to manage: telling the model it *can* improvise seems to confuse it a bit, and codebook coverage drops.

Four codebook strategies:

1. Stick only to the codebook. Anything that doesn't match is dropped.
2. Stick to the codebook mostly, but let the AI code other things too. Tell it to flag the new ones with a tag like `[new]` or a trailing `*`, so you can find and review them later.
3. Compromise. Use only top-level labels from the codebook, but let the AI improvise the second part of a hierarchical label. See [Hierarchical coding.](#)]
4. Free coding

Related to the above...

Label style: *In vivo* or abstract

When free coding, you can ask for *in vivo* labels (close to the original text) or for more abstract labels. There are many ways to instruct: "talk like a social scientist" or "talk like a local newspaper editor".

You can suggest to the AI to use a common “social sciences” vocabulary like:

- presence of resources
- lack of resources
- more/better income
- more/better motivation
- more/better support from peers

W3lcom3!

Label style: Using tags

There's often more to a coding task than just labels. *Tags* are short bits of text in brackets attached to labels: **stressed patients (before surgery)**, **stressed patients (after surgery)**. Tags help you build up a system of labels from smaller parts.

Label style: Opposites

See [!Opposites and sentiment in AI coding](#) and [Opposites](#).

Label style: Hierarchical

You can impose hierarchical labels even when free coding. See [Hierarchical coding](#).

Columns

We also often ask for *columns* unrelated to the labels, such as a sentiment assessment. Columns can be useful for any systematic attribute you can code consistently across most links. Note the difference: columns code attributes of *links*; tags code attributes of *factors*.

Asking the model to fill extra columns alongside coding is extra work for it. Expect either coverage (links found) or precision to drop.

Columns: Sentiment

Zero-codebook coding usually leads on to magnetic soft recoding, and in embedding space **decrease in X** sits close to **increase in X**.

With an explicit codebook, you can get round this by (maybe) suggesting both sides of any factor likely to appear positively and negatively, e.g. **increased income** and **decreased income**.

When you don't specify a codebook, get the model to code sentiment for each link.

Sentiment is a kind of column.

A sentiment column lets you tell them apart.

Sentiment can also be interesting for other reasons.

Other things which are important in the prompt

Often you want to tell the model what the work is for: the context, named entities, the audience. We *iterate* on the prompt, trying versions and comparing results.

Context

...

Named entities

We want the coding AI to recognise words or phrases or abbreviations specific to our project which are not general knowledge and also know that there may be different words for the same thing, e.g. different ways of talking about the same project or organisation; and we usually want it to then just use one preferred phrase even if there are alternatives in the text.

Our preferred phrase should normally be in the same language as the other labels we ask it to make.

Coding style: *holistic* or *claim-by-claim*?

Two main approaches.

Holistic: we ask the model for an overall diagram of the causal network in the current chunk of text². The model decides what the main links are, but we still ask for a quote behind each one.

Especially when using a holistic style, it can be useful to add a second iteration to “mop up” anything missed.

Claim by claim: we ask the model to find each individual causal link. Hundreds of tweaks and heuristics later, this style still struggles to tell a connected story even when the text contains one. Suppose the text supports A to B to C to D, but the model lazily codes A to B and C to D, using slightly different labels for B and C. Claim-by-claim coding only really works if you plan to recode afterwards: you hope a later recoding pass spots that B and C mean the same thing and rejoins the chain.

Model

For relatively straightforward cases, newer or bigger models are not necessarily better for coding. We have had good results with Gemini Flash, for example.

How many iterations?

Additional iterations can be useful:

- For checking and improving quality
- For increasing coverage (finding more actual causal claims)
- For adding additional information (e.g. more columns)

Multi-stage prompts (separated by **====**) let you split coding into sequential passes. The UI does this for you. Mechanics in [!IOIO-Auto-coding](#) and [AI coding](#) under "Prompt sections".

If you are asking the AI to provide substantial additional coding, such as adding more columns like, say, "significance" or "certainty" or to add a translation or some other text column, we recommend using one or more additional iterations for this because you don't want the AI to miss out or mis-code the actual links, which after all are the most important output of the whole process.

More advanced models are more capable of doing more things at the same time during coding.

Chunk and sample strategy

If a single source is short, you can map the whole thing in one go. More often you have either much longer sources, which the app breaks into chunks for you, or many sources.

Recall and precision

A pretty hard rule: the more text you give the model at once, the less dense its coding gets. One page might yield 20 links; 5 pages might also yield 20 links. Sometimes the 20 it picks out of 5 pages really are the most important, but this is not certain, and you are leaving the model to decide. Often you just want better recall, and that means smaller chunks. Don't give the model too much freedom to decide what counts as important.

What does it select

Bigger chunks mean sparser coding. There is a "code everything" setting that does not each source at all: the effect depends on how long your sources are. Otherwise, work in smaller chunks. Aiming to pick up *every* causal claim is unrealistic, but the smaller the chunk, the more you'll catch.

Recoding style

Once the first coding run exists, the next question is how to deal with overlapping labels. See [Different kinds of coding and recoding](#)

Recoding from scratch means arriving at a clean revised codebook and then recoding everything from the beginning. That's more expensive and slower, but usually gives better results than magnetically recoded labels alone (see [Different kinds of coding and recoding](#)).

- *Hard recoding*: revise the codebook and code again.
- *Links recoding*: use AI Answers / Links to recode links, with quote and context available.
- *Factors recoding*: use AI Answers / Factors to recode factor labels directly.
- *Soft recoding*: use clustering or magnetic labels as a softer recoding layer.

Related

- [chapter intro](#)
- [AI coding](#): app docs for the AI Coding panel
- [AI answers panel](#): app docs for AI Answers
- [!010- Auto-coding](#): older detailed notes on the auto-coding prompt
- [!Opposites and sentiment in AI coding](#)
- [! Autoclustering with embeddings](#)
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- [AI in evaluation actually show your working!](#): the transparency argument
- [Just add rigour Three do's and don'ts](#): do's and don'ts for AI text analysis

- [Causal mapping is easy to automate transparently, so is a great fit for scaling with AI](#)

1. Also known as Recall ↩
2. Internally, we ask it for a Mermaid diagram then convert it into a links table. For some reason, LLMs often do a much better job of creating a connected network if you ask for a diagram than if you ask for a set of connected links ↩